

AI Programming Foundations Project

Exploratory analysis of the Titanic passenger dataset

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1 Overview

For this project I used the recommended Titanic training data to build a small, reproducible analysis in Python. My main research question was how survival differed by passenger class, sex, age, and family size. I did not train a predictive model; the purpose was to demonstrate understanding of proper loading, cleaning, summarization, and plotting on a real dataset.

The submission contains the full analysis. It reads the local CSV file using Pandas, applies the cleaning functions, produces grouped summaries, and creates the three figures included below. The repository also contains the source data, the generated figures, and pinned package versions, as well as brief instructions.

2 Dataset Description

I used *train.csv* from Kaggle's Titanic: Machine Learning from Disaster competition. It contains 891 passenger records and 12 columns. The outcome variable, **Survived**, is binary: 342 passengers are recorded as survivors and 549 as non-survivors. Thus, the overall recorded survival rate is 38.4%.

Check	Result	Treatment in the notebook
Missing age	177 rows (19.9%)	Median within recorded sex and class; missingness flag retained
Missing cabin	687 rows (77.1%)	Raw value removed; cabin-known indicator retained
Missing embarkation port	2 rows (0.2%)	Filled with the modal port, S
Exact duplicate rows	0	Checked; no rows removed

Table 1: Data-quality checks performed prior to the analysis.

Important note: The file is a historical passenger list, not a sample designed for statistical inference. The results below describe patterns in these records. They should not be interpreted as causal effects.

3 Workflow Description

The notebook is broken up into six stages:

1. Import required libraries and set display and plotting options.
2. Load CSV and inspect its rows, dimensions, data types, and missingness.
3. Clean column names/values, handle missing data, and create derived fields.
4. Calculate counts and survival rates for the groups used in the analysis.
5. Plot the main comparisons with titles, axis labels, and group sizes.
6. Record findings, assumptions, and limitations.

Two functions perform the cleaning step. `normalize_column_names` converts the original headings to lower-case snake case. `clean_titanic_data` then handles duplicates, whitespace, missing ages, embarkation values, cabin availability, and the derived family-size fields. A third function, `summarize_survival`, returns the passenger count, survivor count, and survival rate for any requested grouping.

4 Key Decisions & Assumptions

All 891 rows were kept, as dropping every incomplete row would discard most of the dataset because cabin is missing for 77.1% of passengers. Instead, I replaced the raw cabin field with a `cabin_known` indicator. This preserves the fact that a cabin was recorded without treating an incomplete cabin string as a reliable location.

Age has less missingness but is still absent for 177 passengers. I imputed those values as the median for the passenger's recorded sex and class. Furthermore, the separate `age_missing` flag distinguishes observed and imputed values. Finally, the two missing embarkation values were filled with the most common port.

Family size is defined as `SibSp + Parch + 1`. This counts the passenger, their siblings/spouse, and their listed parents or children. It does not capture companions outside those relationships, so it is only a proxy for the actual travel group. The cleaned table otherwise follows a consistent structure, with each variable having its own column and each passenger having one row [3].

5 Results & Interpretation

The largest differences appear when survival is grouped by passenger class and sex. Female passengers have a higher survival rate in every class. The rate ranges from 96.8% for first-class women to 13.5% for third-class men. Class still matters within each recorded sex category.

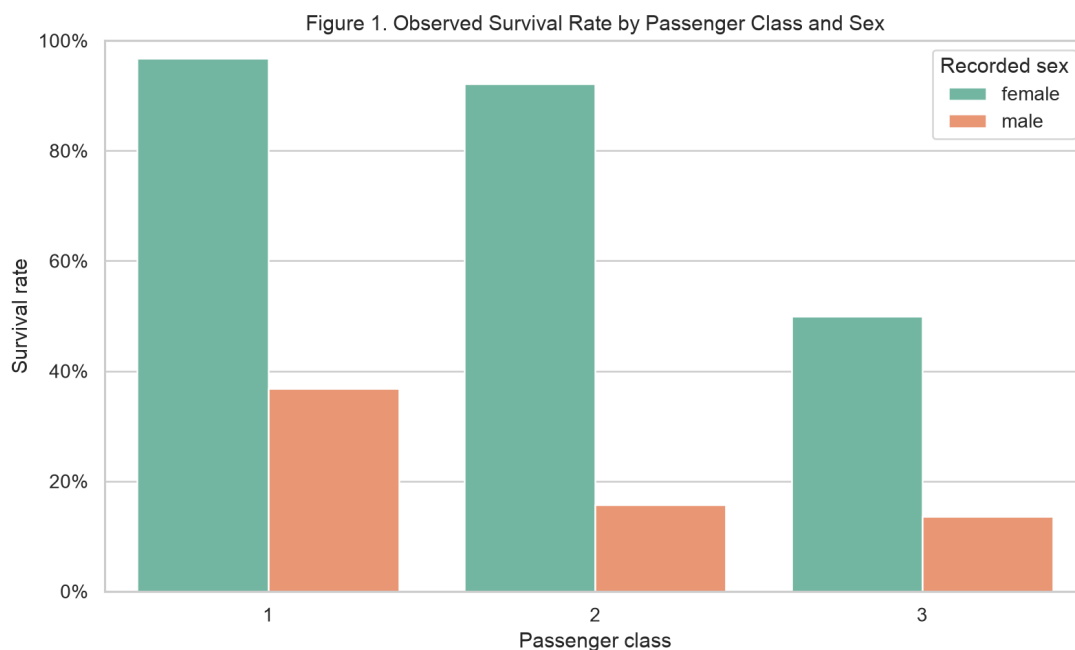


Figure 1: Recorded survival rate by passenger class and sex.

The age distributions overlap substantially. Both outcome groups have a raw median age of approximately 28 years, although the survivor distribution has a visible peak among children. Because 177 ages were filled before plotting, small local differences should be interpreted cautiously.

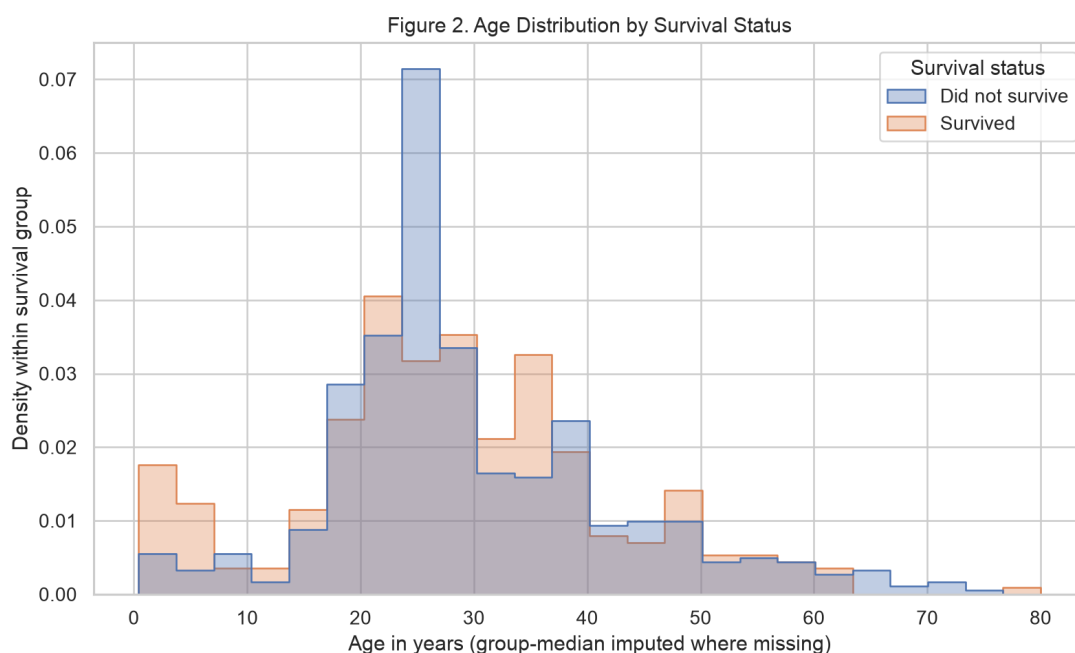


Figure 2: Age distributions for recorded survivors and non-survivors.

Passengers in family groups of two to four have higher recorded survival rates than passengers travelling alone. The relationship is not linear: the rate drops for groups of five or more. Several large-family categories contain very few passengers. For example, family sizes 8 and 11 contain

only 6 and 7 records, so their rates are unstable and should not support broad conclusions.

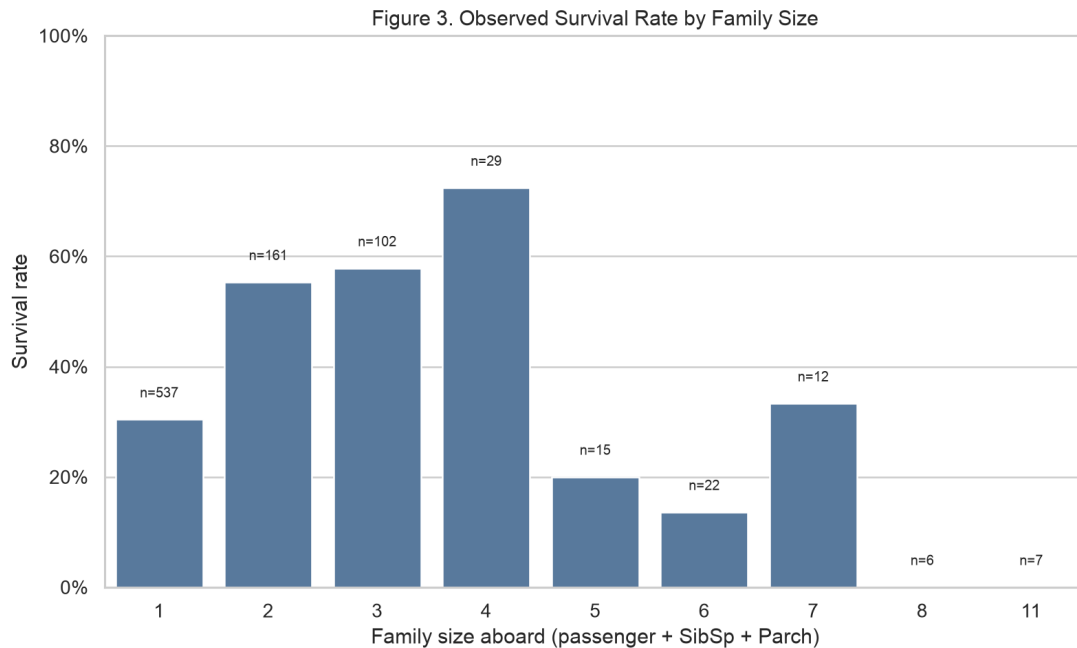


Figure 3: Recorded survival rate by derived family size; labels show group counts.

6 Responsible Practice: Bias & Data Quality

Cleaning choices can change the analysis results and conclusions. A complete-case approach would remove 19.9% of rows because of age alone, while requiring cabin would remove 77.1%. If missingness differs by class or recorded sex, those deletions would also change the comparisons. I therefore kept the rows, recorded missingness explicitly, and reported group counts next to rates.

However, median imputation has its own limitations. It reduces variation and can reinforce the group structure used to calculate the median. A follow-up analysis should compare the current results with complete-case estimates and other imputation methods. The original categories also reflect the conventions and unequal social conditions of the time period. Any predictive use would need separate subgroup checks, a clear reason for using sensitive attributes, and safeguards against data leakage.

7 Reproducibility

The notebook uses relative paths and runs from the repository root. All code cells have been executed without errors. Assertions check the row count, required columns, survival labels, filled age and embarkation values, and the range of the derived family-size field. Package versions are listed in `requirements.txt` and the README explains how to create the environment and

rerun the notebook.

The Git history separates the initial project structure, the executed analysis, and the report. The analysis was developed on a `development` branch and merged into `main`. Keeping the notebook, narrative, environment details, and source data together follows the reproducible teaching approach described by Danchev [1].

8 Sources & Citations

References

- [1] Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, 5(56), 156. <https://doi.org/10.21105/jose.00156>
- [2] Kaggle. (n.d.). *Titanic: Machine Learning from Disaster – Data*. Retrieved 28 September 2026 from <https://www.kaggle.com/c/titanic/data>
- [3] Wickham, H. (2014). Tidy data. *Journal of Statistical Software*, 59(10), 1–23. <https://doi.org/10.18637/jss.v059.i10>